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Dear Mercanis AI Team,

I am applying for the Agentic AI Engineer position. Designing and building agentic systems that automate complex, multi-step workflows — with LLM integration, structured data handling, and reliable fallback behaviour — is the work I have been doing in my Master's thesis and research roles.

My thesis is a production agentic pipeline for automated knowledge graph repair. An autonomous agent (Double DQN, PyTorch) receives structured state from an automotive electronics knowledge graph, selects from a six-action tool schema, calls a GPT-4o mini pipeline via the OpenAI API for structured data extraction, invokes formal OWL reasoning as a validation layer, and iterates until the task resolves. State management, action masking to prevent invalid transitions, retry logic, shaped rewards to penalise loops, and a defined Human-in-the-Loop handoff for edge cases — these are all built-in, not bolted on. The system achieves 88 to 96 percent task success validated on a real dataset of 2,000+ entries. That is not a demo. It is a system I designed, built, evaluated, and own.

On evaluation: I built a quality evaluation workflow to track task success across violation types, catch regressions between training runs, and validate that the system actually improves rather than overfitting to narrow cases. Measuring AI system performance with clear metrics was a first-class concern in this project, not an afterthought.

On LLM integration: I have engineered prompts for consistent, verifiable structured output at scale — pipelines where the LLM output feeds a downstream

formal reasoning system that either accepts or rejects it. Getting that interface reliable under varied inputs required working through hallucination patterns, context window management, and output parsing failures in production conditions.

The knowledge graph dimension of my work is also directly relevant to procurement. Procurement data is inherently structured — suppliers, contracts, categories, constraints, approval chains — and building agents that reason over structured relational state is the core of what I have been doing.

I will be direct about the gaps. I have not used LangGraph, LiteLLM, or Phoenix — my agentic work uses custom orchestration with the raw OpenAI API. I have not deployed to AWS. I have not built RAG pipelines with vector databases, though I understand the architecture from knowledge graph retrieval work and am building toward it. These are real gaps, and I would rather name them.

The foundation I bring is solid: proven agentic system design, LLM integration under real constraints, evaluation discipline, and the habit of owning what I ship. I would welcome the opportunity to discuss how that fits what Mercanis is building.

Best regards,

Resham Joshi